Random Network Routing Evaluation

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Contents

1 Evaluation Setup

Does the relative performance ranking of HEFT, HEFT-1, and HEFT-2 hold on random graphs? We place **50 nodes** uniformly at random in an $L \times L$ square and create bidirectional links between all pairs within $R = 80\,\mathrm{m}$. Varying L controls network density.

Parameter	**Value**
Nodes	50, positions uniform random in $[0, L]^2$
Communication range	$R = 80\,\mathrm{m}$ (`csma_bianchi` PHY rate from distance)
Topology seed	42 (fixed per density level)
Seeds per combo	30 (seeds 1–30)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Routing schemes	W, S, SH, GO, GS, GSD, GSD-D
DAGs	Small (8 tasks, fork-join) and Large (30 tasks, pipeline)
Interference	`csma_bianchi`, 802.11ax 5 GHz 20 MHz

Table 1: Evaluation parameters.

Values reported as mean ± std over 30 seeds. Additional metrics: mean hops (path length), peak link utilization (P-LU), mean transfer duration (XD).

Level	**Side L (m)**	**Links**	**Avg. degree**
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 2: Topology statistics (seed 42).

2 Makespan vs. Density

Each point is the best-routing mean makespan ± std over 30 seeds.

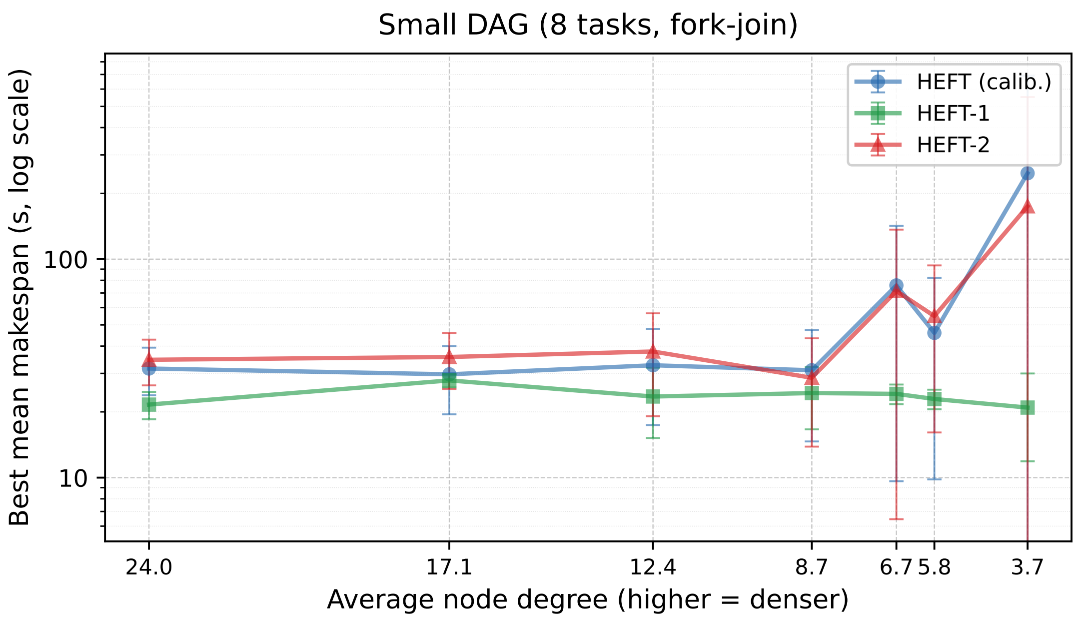

Figure 1: Best-routing makespan vs. avg degree — Small DAG (8 tasks, fork-join). Error bars show ±1 std over 30 seeds.

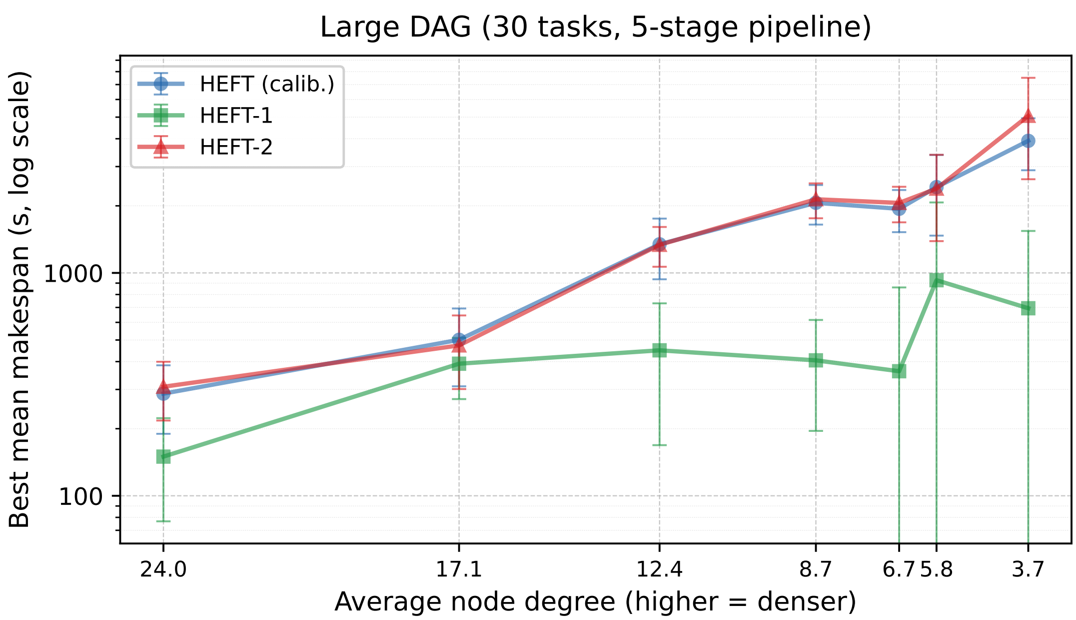

Figure 2: Best-routing makespan vs. avg degree — Large DAG (30 tasks, 5-stage pipeline). Error bars show ±1 std over 30 seeds.

3 Additional Metrics vs. Density (Large DAG)

Computed at the best routing scheme per (density, scheduler) cell.

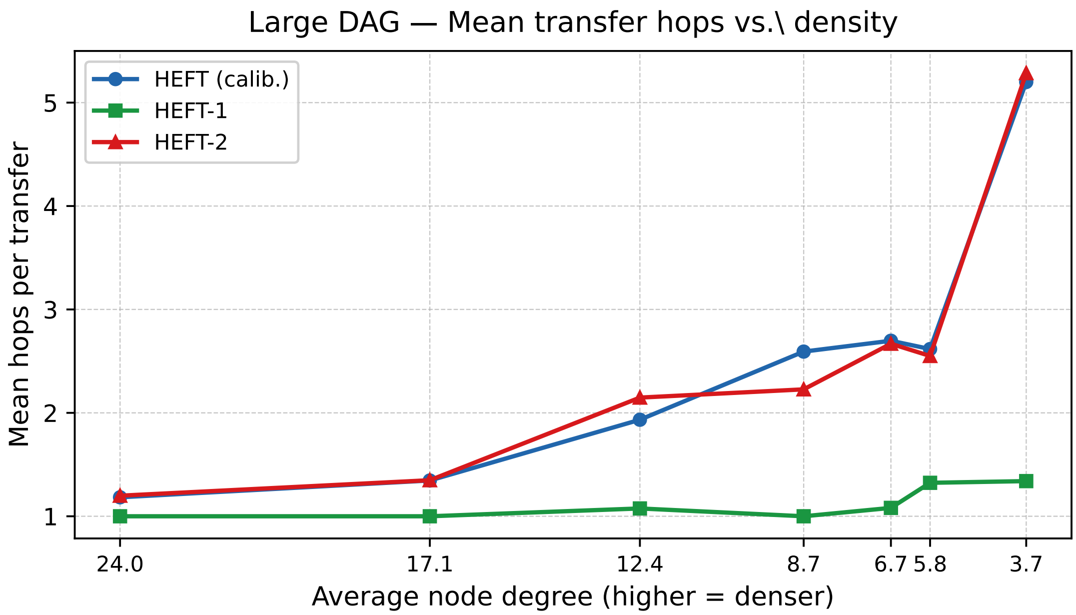

Figure 3: Mean hops per transfer. Values > 1 indicate multi-hop paths.

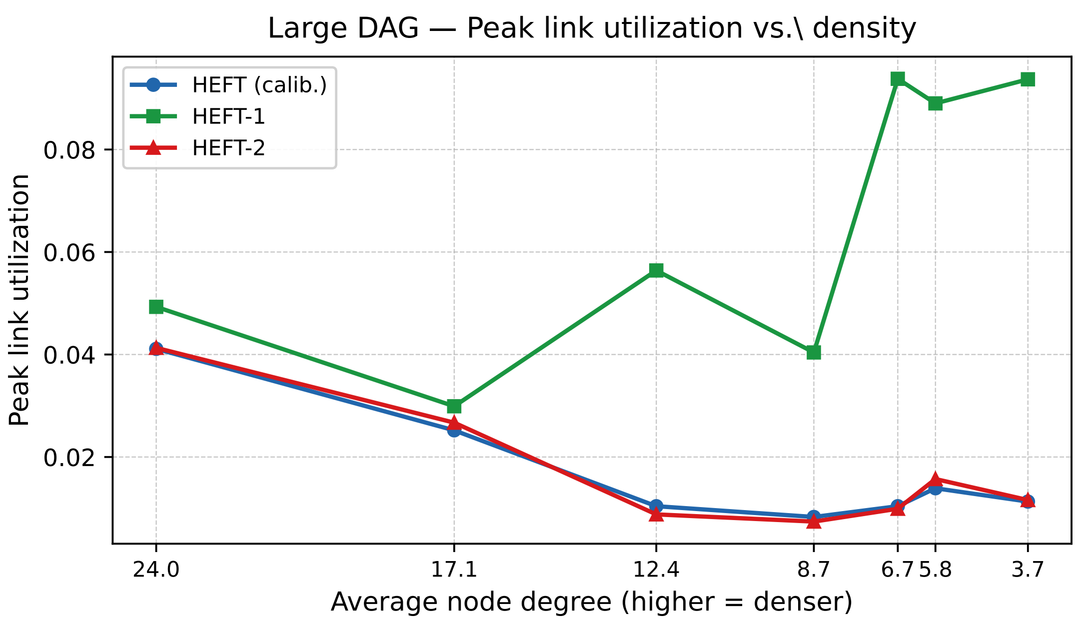

Figure 4: Peak link utilization (max over all links). High values indicate congestion hotspots.

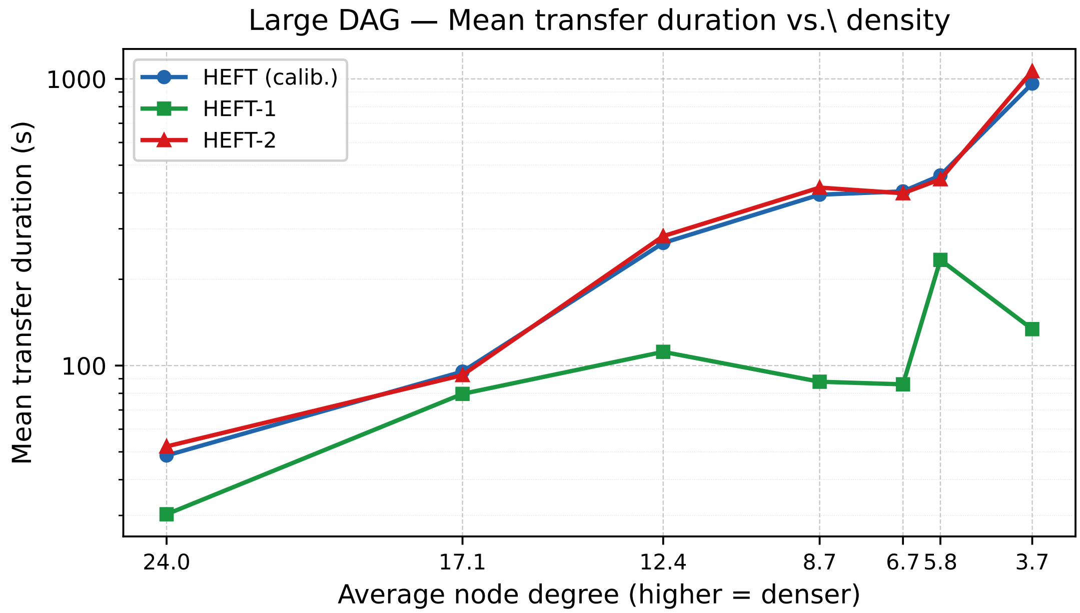

Figure 5: Mean transfer duration (s, log scale). Tracks makespan closely.

4 Best Routing per Density and Scheduler

4.1 Small DAG (8 tasks)

Dens.	Deg.	HEFT (calib.)			HEFT-1			HEFT-2		
		Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU
L150	24.0	S	31.584 ±7.768	1.1/0.20	S	21.594 ±3.096	1.0/0.19	S	34.633 ±8.195	1.1/0.21
L200	17.1	SH	29.712 ±10.225	1.0/0.20	S	27.861 ±2.003	1.0/0.16	SH	35.664 ±10.197	1.0/0.22
L250	12.4	S	32.688 ±15.283	1.1/0.19	S	23.513 ±8.325	1.0/0.17	S	37.796 ±18.701	1.2/0.18
L300	8.7	SH	31.009 ±16.365	1.3/0.18	SH	24.388 ±7.755	1.0/0.18	SH	28.641 ±14.774	1.2/0.17
L350	6.7	SH	75.816 ±66.186	1.8/0.17	SH	24.179 ±2.499	1.0/0.22	SH	71.486 ±65.031	1.7/0.17
L400	5.8	SH	45.980 ±36.165	1.3/0.18	S	22.909 ±2.360	1.0/0.24	SH	54.861 ±38.762	1.4/0.20
L500	3.7	S	247.261±443.820	2.1/0.17	S	**20.926**±9.049	1.0/0.29	GS	174.510±377.184	1.8/0.21

Table 3: Small DAG (8 tasks) — best routing per (density, scheduler). ms = mean ± std makespan (s); H = mean hops; PLU = peak link util. **Bold green**: overall best.

4.2 Large DAG (30 tasks)

Dens.	Deg.	HEFT (calib.)			HEFT-1			HEFT-2		
		Rt	ms (s)	H/PLU	Rt	ms (s)	H/PLU	Rt	ms (s)	H/P
L150	24.0	SH	287.323 ±97.615	1.2/0.04	SH	**149.899**±73.082	1.0/0.05	SH	308.494 ±90.917	1.2/0
L200	17.1	SH	501.174 ±191.253	1.3/0.03	SH	391.400 ±119.854	1.0/0.03	SH	472.856 ±171.599	1.3/0
L250	12.4	SH	1344.549±408.317	1.9/0.01	S	449.477 ±280.738	1.1/0.06	S	1336.595±270.894	2.1/0
L300	8.7	S	2063.615±416.010	2.6/0.01	SH	405.276 ±209.779	1.0/0.04	SH	2140.430±381.959	2.2/0
L350	6.7	S	1939.103±415.676	2.7/0.01	SH	362.181 ±498.494	1.1/0.09	S	2060.780±374.358	2.7/0
L400	5.8	SH	2426.675±956.493	2.6/0.01	W	927.487 ±1143.686	1.3/0.09	SH	2388.566±999.895	2.6/0
L500	3.7	S	3914.460±1026.203	5.2/0.01	GSD-D	694.194 ±848.372	1.3/0.09	GSD-D	5068.567±2436.577	5.3/0

Table 4: Large DAG (30 tasks) — best routing per (density, scheduler). ms = mean ± std makespan (s); H = mean hops; PLU = peak link util. **Bold green**: overall best.

5 Full Results Tables

Mean ± std makespan (s), mean hops, peak link util over 30 seeds. **Bold green**: overall best for density+DAG. Red: worst per scheduler column.

5.1 L150 (avg. degree 24.0) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	494.056±395.759	4.0	0.017	43.651±21.247	2.2	0.108	520.163±438.020	4.3	0.018
S	31.584±7.768	1.1	0.198	**21.594**±3.096	1.0	0.187	34.633±8.195	1.1	0.211
SH	36.964±9.209	1.0	0.208	21.956±3.415	1.0	0.187	35.639±9.482	1.0	0.204
GO	536.541±264.293	3.6	0.044	242.145±16.366	3.8	0.059	612.818±246.323	3.6	0.036
GS	589.876±180.502	3.7	0.040	286.759±82.371	3.4	0.059	562.633±188.999	3.8	0.042
GSD	640.136±286.646	4.3	0.038	286.083±111.159	2.0	0.056	639.593±293.681	4.4	0.040
GSD-D	414.070±86.841	4.7	0.058	144.089±33.607	2.0	0.095	409.654±120.001	4.9	0.063

Table 5: L150 ($L = 150\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.2 L200 (avg. degree 17.1) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	517.000±532.748	3.8	0.076	43.892±24.800	1.9	0.127	460.683±495.829	3.6	0.083
S	38.313±21.088	1.3	0.158	**27.861**±2.003	1.0	0.164	38.134±19.738	1.3	0.171
SH	29.712±10.225	1.0	0.200	28.909±2.398	1.0	0.190	35.664±10.197	1.0	0.217
GO	872.239±1243.295	3.5	0.066	411.368±399.410	3.1	0.068	1184.579±1483.464	3.5	0.062
GS	1088.416±1411.899	3.3	0.060	489.974±469.038	3.1	0.057	888.119±1236.443	3.4	0.067
GSD	990.931±1329.407	3.0	0.052	483.804±470.428	2.2	0.053	936.577±1214.990	4.7	0.053
GSD-D	459.646±725.769	3.9	0.115	229.148±212.339	1.8	0.109	830.506±1079.304	3.8	0.091

Table 6: L200 ($L = 200\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.3 L250 (avg. degree 12.4) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	609.587±762.389	2.8	0.116	29.521±17.147	1.1	0.165	546.569±737.574	2.7	0.116
S	32.688±15.283	1.1	0.192	**23.513**±8.325	1.0	0.170	37.796±18.701	1.2	0.180
SH	41.687±19.984	1.3	0.172	25.843±9.323	1.0	0.174	38.534±19.627	1.2	0.170
GO	596.181±852.593	2.9	0.086	106.264±96.645	2.5	0.111	405.795±704.200	2.7	0.101
GS	250.967±285.347	2.7	0.106	91.428±81.701	2.4	0.119	314.147±287.965	2.8	0.093
GSD	468.916±538.406	2.6	0.088	118.211±97.194	1.1	0.115	629.508±591.167	3.1	0.075
GSD-D	190.333±149.557	2.5	0.116	84.703±76.800	1.1	0.132	209.977±177.291	3.1	0.101

Table 7: L250 ($L = 250\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.4 L300 (avg. degree 8.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	847.135±960.462	3.4	0.106	28.512±8.633	1.1	0.178	720.078±934.033	3.1	0.120
S	34.708±17.786	1.4	0.166	26.982±8.666	1.0	0.181	33.490±17.405	1.3	0.168
SH	31.009±16.365	1.3	0.181	**24.388**±7.755	1.0	0.175	28.641±14.774	1.2	0.171
GO	1067.400±1275.536	2.5	0.072	151.303±139.010	2.3	0.112	730.413±1128.691	2.3	0.083
GS	916.827±1177.032	2.6	0.071	129.153±136.506	2.1	0.122	576.451±891.467	2.3	0.083
GSD	361.737±647.347	2.1	0.116	109.158±128.730	1.2	0.133	871.740±845.597	3.8	0.073
GSD-D	476.092±792.718	2.4	0.127	116.086±115.703	1.3	0.156	475.722±702.279	2.7	0.119

Table 8: L300 ($L = 300\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.5 L350 (avg. degree 6.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	692.048±1028.635	2.8	0.111	24.745±6.684	1.1	0.233	985.791±1112.765	3.4	0.091
S	414.203±830.766	2.2	0.141	25.020±5.279	1.1	0.211	331.440±761.427	1.9	0.153
SH	75.816±66.186	1.8	0.169	**24.179**±2.499	1.0	0.217	71.486±65.031	1.7	0.167
GO	1115.620±1443.721	2.3	0.094	40.942±23.034	1.6	0.189	1312.776±1483.877	2.5	0.084
GS	1015.965±1413.352	2.2	0.099	52.288±22.199	1.8	0.170	625.956±1197.779	1.9	0.115
GSD	336.893±896.596	1.9	0.123	54.115±21.714	1.2	0.168	1215.382±1466.386	3.2	0.088
GSD-D	1113.997±1444.052	3.0	0.103	39.763±23.657	1.1	0.223	917.628±1373.200	2.7	0.113

Table 9: L350 ($L = 350\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.6 L400 (avg. degree 5.8) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	837.123±1083.421	3.1	0.138	29.321±8.893	1.3	0.209	689.413±1030.510	2.7	0.144
S	615.176±994.768	2.6	0.139	**22.909**±2.360	1.0	0.238	615.634±994.487	2.6	0.149
SH	45.980±36.165	1.3	0.181	23.741±2.347	1.0	0.231	54.861±38.762	1.4	0.203
GO	503.103±883.794	2.7	0.108	207.313±170.499	2.1	0.118	512.725±879.083	2.8	0.104
GS	440.730±819.410	2.6	0.106	255.294±138.328	2.5	0.086	384.038±743.134	2.6	0.119
GSD	437.806±820.867	2.8	0.122	243.985±155.792	1.3	0.086	592.820±924.366	3.2	0.109
GSD-D	423.879±825.860	2.3	0.138	117.097±115.572	1.1	0.182	433.288±822.376	2.7	0.137

Table 10: L400 ($L = 400\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.7 L500 (avg. degree 3.7) — Small DAG (8 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	356.336±508.470	2.9	0.168	21.669±8.205	1.0	0.282	210.881±413.512	2.0	0.197
S	247.261±443.820	2.1	0.174	**20.926**±9.049	1.0	0.286	392.711±523.049	2.9	0.149
SH	501.809±549.817	3.4	0.121	23.151±8.785	1.0	0.297	465.443±543.565	3.2	0.131
GO	501.795±549.830	3.5	0.134	22.127±7.384	1.0	0.277	465.438±543.568	3.5	0.135
GS	429.050±534.708	3.2	0.167	21.384±8.354	1.0	0.282	174.510±377.184	1.8	0.211
GSD	392.711±523.049	3.1	0.149	32.377±21.924	1.0	0.208	356.363±508.452	2.8	0.141
GSD-D	318.912±491.322	2.6	0.173	21.863±8.663	1.0	0.312	318.765±491.411	2.6	0.198

Table 11: L500 ($L = 500\,\mathrm{m}$) — Small DAG (8 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.8 L150 (avg. degree 24.0) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1530.829±223.415	5.5	0.007	542.056±310.803	2.5	0.020	1504.689±235.408	5.6	0.007
S	715.431±279.178	1.7	0.020	198.784±101.554	1.2	0.047	675.535±308.317	1.7	0.022
SH	287.323±97.615	1.2	0.041	**149.899**±73.082	1.0	0.049	308.494±90.917	1.2	0.041
GO	3540.427±676.048	3.8	0.008	2117.429±762.023	3.5	0.021	3383.308±672.062	3.8	0.009
GS	3567.396±735.216	3.9	0.008	2501.810±691.324	3.7	0.016	3589.212±705.323	3.8	0.008
GSD	3256.273±422.823	5.5	0.008	2192.774±755.378	2.5	0.018	3309.441±386.379	5.5	0.008
GSD-D	2601.980±292.511	5.4	0.010	1805.307±545.149	2.6	0.021	2568.484±279.133	5.5	0.011

Table 12: L150 ($L = 150\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.9 L200 (avg. degree 17.1) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1943.409±388.855	5.7	0.007	1079.533±492.574	2.5	0.016	1822.756±382.398	5.4	0.007
S	831.126±115.776	1.8	0.016	410.010±137.880	1.0	0.030	860.765±153.697	1.8	0.015
SH	501.174±191.253	1.3	0.025	**391.400**±119.854	1.0	0.030	472.856±171.599	1.3	0.027
GO	4049.688±600.687	3.7	0.010	3425.021±1261.481	3.6	0.015	4131.985±457.422	3.7	0.010
GS	3973.898±815.731	3.7	0.010	4263.847±2151.727	3.5	0.011	3671.203±731.394	3.7	0.010
GSD	3531.402±676.131	5.6	0.010	3902.399±2000.261	2.5	0.013	3652.316±735.637	5.7	0.010
GSD-D	3131.900±861.145	5.5	0.013	2520.026±472.214	2.3	0.018	2981.828±838.339	5.7	0.013

Table 13: L200 ($L = 200\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.10 L250 (avg. degree 12.4) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1812.396±310.729	4.8	0.009	626.690±278.479	1.9	0.032	1852.661±281.293	4.7	0.009
S	1357.554±260.999	2.1	0.008	**449.477**±280.738	1.1	0.056	1336.595±270.894	2.1	0.009
SH	1344.549±408.317	1.9	0.010	559.928±276.236	1.0	0.044	1361.136±392.196	1.9	0.010
GO	3063.776±440.036	3.7	0.009	1262.239±297.262	3.1	0.029	2986.991±407.257	3.7	0.008
GS	3048.733±495.837	3.4	0.007	1820.238±996.768	3.0	0.025	3023.702±470.928	3.5	0.008
GSD	2530.877±381.148	4.8	0.008	1835.618±882.414	2.0	0.019	2602.391±333.341	4.8	0.008
GSD-D	1951.639±313.426	4.7	0.011	1226.291±669.369	1.9	0.029	1924.656±320.437	4.7	0.011

Table 14: L250 ($L = 250\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.11 L300 (avg. degree 8.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2639.992±634.022	4.9	0.006	809.067±640.877	1.9	0.047	2726.267±613.490	5.5	0.006
S	2063.615±416.010	2.6	0.008	428.149±248.501	1.1	0.041	2156.510±308.769	2.7	0.007
SH	2233.182±362.908	2.4	0.007	**405.276**±209.779	1.0	0.040	2140.430±381.959	2.2	0.007
GO	2810.638±419.397	4.0	0.008	1078.668±370.577	1.9	0.033	2859.602±427.744	4.0	0.008
GS	2822.218±476.802	3.9	0.009	1027.168±284.955	1.9	0.040	2807.280±538.642	3.9	0.009
GSD	2769.135±245.327	5.2	0.008	1471.472±611.280	1.9	0.022	2725.680±277.762	5.1	0.008
GSD-D	2740.669±815.114	5.6	0.009	998.452±50.359	1.8	0.035	2500.433±927.019	5.2	0.010

Table 15: L300 ($L = 300\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.12 L350 (avg. degree 6.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2262.481±311.049	4.3	0.009	627.582±774.416	1.3	0.093	2378.291±263.160	4.3	0.009
S	1939.103±415.676	2.7	0.010	767.926±688.464	1.3	0.071	2060.780±374.358	2.7	0.010
SH	2142.480±533.937	2.5	0.011	**362.181**±498.494	1.1	0.094	2155.497±573.914	2.6	0.011
GO	2577.160±855.336	3.6	0.010	880.890±773.572	1.7	0.050	2406.193±763.198	3.4	0.011
GS	2426.125±701.306	3.9	0.010	1185.292±877.412	1.8	0.041	2483.959±651.898	3.7	0.012
GSD	2625.803±747.668	3.9	0.010	1028.508±734.175	1.3	0.036	2454.205±730.232	4.0	0.010
GSD-D	2575.997±962.700	4.4	0.011	891.407±751.442	1.3	0.050	2353.488±909.641	4.1	0.011

Table 16: L350 ($L = 350\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.13 L400 (avg. degree 5.8) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2844.226±831.929	3.6	0.006	**927.487**±1143.686	1.3	0.089	2980.537±784.045	4.4	0.006
S	2947.258±813.232	3.8	0.007	1080.936±1271.055	1.1	0.067	2919.019±788.194	4.3	0.007
SH	2426.675±956.493	2.6	0.014	1252.065±1325.088	1.1	0.064	2388.566±999.895	2.6	0.016
GO	3331.649±276.965	4.1	0.009	1229.818±1365.355	1.6	0.072	3336.711±299.362	4.1	0.008
GS	3154.020±318.127	4.7	0.009	1611.463±1425.890	1.6	0.063	3239.241±352.323	4.5	0.009
GSD	3209.028±430.747	3.9	0.008	1621.605±1543.986	1.3	0.035	3065.852±370.610	4.4	0.007
GSD-D	3070.476±683.704	3.8	0.008	1202.449±1351.971	1.3	0.068	3205.536±605.676	4.3	0.007

Table 17: L400 ($L = 400\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

5.14 L500 (avg. degree 3.7) — Large DAG (30 tasks)

Route	HEFT			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	5494.190±2715.245	5.6	0.010	906.925±939.040	1.3	0.097	5732.305±2309.919	5.0	0.011
S	3914.460±1026.203	5.2	0.011	792.757±910.573	1.0	0.108	6166.377±2844.477	4.9	0.011
SH	5924.170±2780.786	5.0	0.011	1035.847±972.957	1.1	0.091	6870.982±2578.797	4.7	0.011
GO	4771.168±1976.106	5.3	0.012	923.603±961.934	1.2	0.100	5773.163±2438.199	4.6	0.013
GS	4544.688±1723.611	5.5	0.011	860.593±945.260	1.2	0.104	5767.751±2680.004	4.8	0.012
GSD	5560.900±2387.494	4.7	0.013	938.632±953.460	1.2	0.086	5386.756±1930.721	4.7	0.013
GSD-D	5212.339±2344.961	5.1	0.012	**694.194**±848.372	1.3	0.094	5068.567±2436.577	5.3	0.012

Table 18: L500 ($L = 500\,\mathrm{m}$) — Large DAG (30 tasks). ms = mean±std; H = mean hops; PLU = peak link util.

6 Analysis

6.1 HEFT-1 Dominance Is Robust Across All Densities

HEFT-1 wins at every density level for both DAGs. For the large DAG, the advantage over calibrated HEFT ranges from 1.9× at L150 to 5.6× at L500. The standard deviations under HEFT and HEFT-2 are large (often > 50% of the mean) because csma_bianchi interference is highly sensitive to the exact traffic pattern, which varies across seeds. HEFT-1's std is much smaller: co-location eliminates most transfers, leaving only compute variance.

6.2 SH vs. S: Density-Dependent Reversal

Under HEFT-1, SH beats S at L150–L300 for the large DAG (mean-hops = 1.0 for both, confirming co-location). The routing tie-break wins come from how each scheme selects the single hop: SH prefers the nearest (lowest-loss) link, reducing PHY-layer retransmission and thus interference variance. At sparse topologies (L400–L500), fewer 1-hop options exist and S regains the lead by picking the highest-bandwidth available link.

6.3 Hops, Peak Link Util., and Transfer Duration

The additional metrics confirm that routing choice under csma_bianchi is primarily about *which* 1-hop link is used, not multi-hop avoidance:

- Mean hops is ≈ 1.0 for HEFT-1 across all density levels, confirming that co-located tasks transfer directly to adjacent nodes.
- Calibrated HEFT and HEFT-2 use 2–4 hops at high density (L150–L200), where tasks are spread across many nodes. Multi-hop paths under csma_bianchi amplify interference—each relay adds an active transmitter—explaining the large makespan gap.
- Peak link utilization under HEFT-1 is low (< 0.06) because co-located tasks do not use links; the few transfers that occur are short and non-overlapping. Under HEFT/HEFT-2 peak link util can exceed 0.9, indicating severe bottleneck congestion.

6.4 Reproducing These Results

```
cd ncsim/
python run_random_eval.py                  # ~3500 s (8820 runs)
python compute_interference_metrics.py      # reads traces, ~3 min
python gen_random_network_report.py         # generates tex + plots
cd docs/ && pdflatex random_network_results.tex
```